

This unit is assessed by means of a written examination paper of 1 hour 35 minutes duration. The paper will consist of objective, short-answer and long-answer questions. Students may be required to apply their knowledge and understanding of physics to situations that they have not encountered before. The total number of marks available for this examination paper is 80. It contributes 40% to IA2 and 20% to the IAL in Physics.

It is recommended that students have access to a scientific calculator for this paper.

Students will be provided with the formulae sheet shown in *Appendix 5*. Any other physics formulae that are required will be stated in the question paper.

The quality of written communication will be assessed in the context of this unit through questions which are labelled with an asterisk (*). Students should take particular care with spelling, punctuation and grammar, as well as the clarity of expression, on these questions.

4.3 Thermal energy

This topic covers specific heat capacity, internal energy and the ideal gas equation.

This topic may be taught using applications that relate to, for example, space technology.

Students will be assessed on their ability to:		Suggested experiments
109	investigate, recognise and use the expression $\Delta E = mc\Delta\theta$	Measure specific heat capacity of a solid and a liquid using, for example, temperature sensor and data logger
110	explain the concept of internal energy as the random distribution of potential and kinetic energy amongst molecules	
111	explain the concept of absolute zero and how the average kinetic energy of molecules is related to the absolute temperature	
112	recognise and use the expression $\frac{1}{2}m\langle c^2 \rangle = \frac{3}{2}kT$	
113	use the expression $pV = NkT$ as the equation of state for an ideal gas	Use temperature and pressure sensors to investigate the relationship between p and T Experimental investigation of the relationship between p and V